

# CoRoT-1b

R-2631

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_CoRoT-1\_141203 · created 2026-08-29T00:27:56+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2456994.802 +/- 0.0027 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.104 +/- 0.041                |
| Transit depth (Rp/Rs)^2                           | 1.08 +/- 0.84 [%]              |
| Orbital Inclination (inc)                         | 85.13 +/- 2.83                 |
| Airmass coefficient 1 (a1)                        | 0.6 +/- 0.39                   |
| Airmass coefficient 2 (a2)                        | 0.44 +/- 0.38                  |
| Scatter in the residuals of the lightcurve fit is | 74.91 %                        |
| Best Comparison Star                              | None                           |
| Optimal Aperture                                  | 3.47                           |
| Optimal Annulus                                   | 4.12                           |
| Transit Duration (day)                            | 0.099 +/- 0.013                |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.104 ± 0.041 vs 0.1388 (deviation 0.85σ) |
| Duration fit vs published | 0.099 d vs 0.1046 d (deviation 0.43σ)     |
| Mid-time O–C              | -4.7 min                                  |
| Depth SNR                 | 2.5                                       |
| Residual scatter          | 74.91 %                                   |

### Light curve

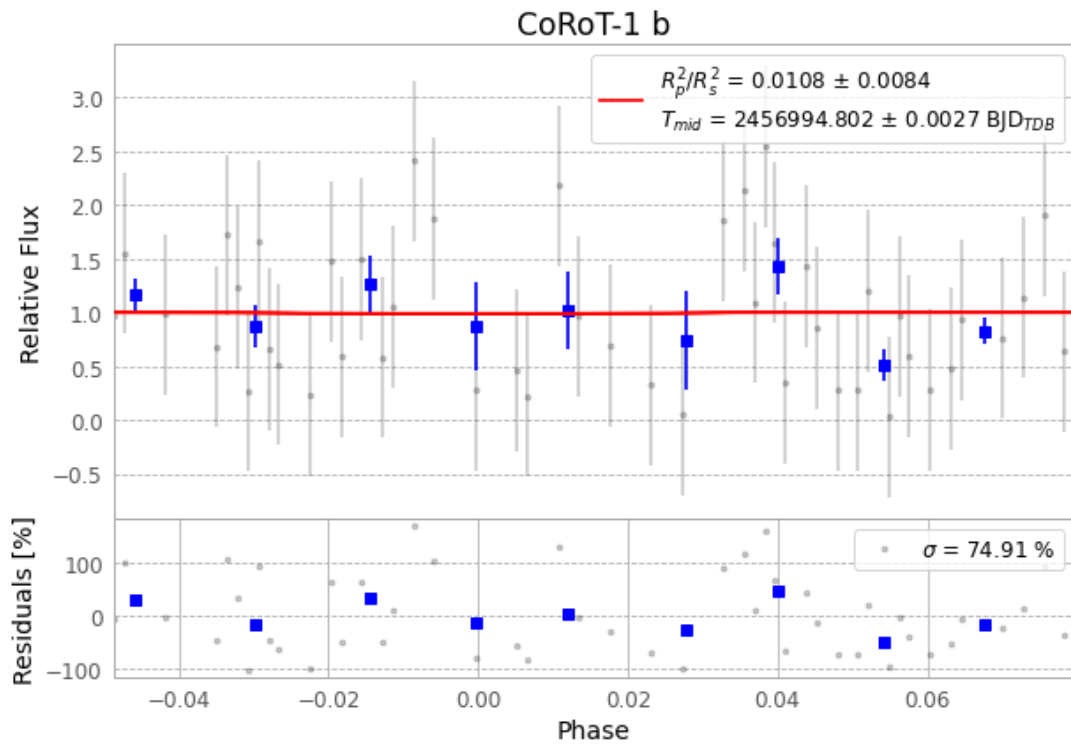


Figure 1 — FinalLightCurve\_CoRoT-1 b\_2014-12-02.png (SHA-256 7ca7bc0d98c6b6e6...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [79, 150], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 98b56e294386f6bf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

94 FITS frames (62 MB), CoRoT-1141203052112.FITS → CoRoT-1141203100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-141203.log (442bd2bd7b0b...), FinalLightCurve\_CoRoT-1 b\_2014-12-02.png (7ca7bc0d98c6...), FinalLightCurve\_CoRoT-1 b\_2014-12-02.csv (541d72e0a3cc...)

## Compute resources

Wall time 179.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:27Z.